

Spencer Wozniak

586-522-6021 | spencer@serelora.com | [LinkedIn](#) | [Website](#) | [GitHub](#) | [Google Scholar](#)

I am an entrepreneur and software engineer focused on applied AI and complex data systems. My background spans full-stack engineering, AI research, and clinical operations, with experience in emergency, inpatient, and behavioral healthcare. Grounded in first principles, I build systems that empower people to serve others.

Experience

Co-Founder & CTO

June 2025 - Present

Serelora (www.serelora.com) | San Diego, CA

- Founded and led a healthcare SaaS company, initially delivering B2B CRM and workflow automation tools before pivoting to AI-powered EHR systems after identifying clinical documentation and interoperability gaps through customer outreach.
- Designed an FHIR-native, AI-first EHR architecture emphasizing traceability and clinician trust.
- Modeled longitudinal patient data in PostgreSQL using FHIR-aligned schemas, enabling structured relationships across clinical history, documentation, and operational workflows.
- Architected a manager-orchestrated system of domain-specific clinical AI agents, dynamically routing requests to labs, medications, and documents agents with scoped retrieval and context awareness.
- Designed agent workflows to balance explainability, performance, and clinical constraints.
- Implemented HIPAA-aligned technical safeguards, including RBAC, audit logging, access boundaries.
- Worked extensively with incomplete, inconsistent, and multi-source clinical data, designing ingestion and reconciliation logic resilient to real-world documentation variability.
- Worked with healthcare organizations to translate workflows into concrete product requirements.
- Led ongoing rollout and support with clients, coordinating calls and check-ins to ensure systems functioned reliably in live environments and adapted rapidly as needs evolved.
- Led go-to-market efforts including outreach through cold email, LinkedIn, and in-person meetings; as well as discovery calls, on-site demos, and deal negotiations with healthcare organizations.

Founder & Lead Tutor

November 2024 – December 2025

WozPrep (www.wozprep.org) | San Diego, CA

- Founded and operated a private tutoring service supported by a custom-built web application.
- Developed a student portal to manage inquiries, host content, and deliver MCAT practice materials.
- Built a web-based testing and question bank system featuring original, AAMC-style questions.
- Improved student outcomes by up to 50 percentile points through individualized tutoring.
- Applied Socratic method and positive psychology to help critical thinking, motivation, and confidence.
- Designed personalized study plans based on student performance and time constraints.

Co-Founder & Lead Engineer

May 2023 – August 2023

Clinical Training Platform for IMGs | Rochester, MI

- Co-founded and led engineering for a clinical training platform preparing international medical graduates (IMGs) for U.S. hospital rotations and residency.
- Built a full-stack web application delivering case-based clinical education, SOAP note training, and AI-assisted feedback.
- Built a React + TypeScript frontend (Vite) supporting authenticated users, course purchase flows, interactive case navigation, video-based learning, and real-time note-taking.
- Developed a Python (Flask) backend handling user authentication, course progress tracking, session management, and secure data persistence.
- Built an AI-powered tutor to provide context-aware feedback on HPI writing and clinical reasoning.
- Designed prompt-engineering workflows and integrated text-to-speech pipelines for AI feedback delivery, enabling multimodal learning experiences.
- Collaborated with physician educators to translate real hospital cases into interactive modules.
- Supported early pilot deployments with medical students and IMG cohorts; platform used for demonstrations to hospital leadership and international partners.

Research Experience

Research Assistant

September 2020 – April 2025

Biochemistry Department, Michigan State University | East Lansing, MI

- Conducted multi-year research applying graph-based representations to model molecular systems.
- Developed and evaluated transformer-based graph neural networks (GNNs) to predict molecular properties, leveraging message passing over local and global structural neighborhoods.
- Built data pipelines in Python and C++ to transform raw molecular dynamics outputs into graph-structured datasets suitable for analysis and training models.
- Curated, validated, and maintained large datasets for training and benchmarking AI models.
- Collaborated with interdisciplinary teams to interpret results and refine modeling assumptions.

Publications

- Wozniak S, Janson G, Feig M. Accurate Predictions of Molecular Properties of Proteins via Graph Neural Networks and Transfer Learning. *Journal of Chemical Theory and Computation*. 2025. <https://doi.org/10.1021/acs.jctc.4c01682>
- Wozniak S, Feig M. Diffusion and Viscosity in Mixed Protein Solutions. *The Journal of Physical Chemistry B*. 2024. <https://doi.org/10.1021/acs.jpcb.4c06877>

Clinical Experience

Applied Behavior Analysis Therapist

August 2024 – December 2025

Coyne and Associates | San Diego, CA

- Conducted individualized therapy sessions with children diagnosed with developmental disabilities.
- Collected data in an EHR to track client progress, analyze behavioral trends, and refine interventions.
- Trained and supervised new therapists, providing structured feedback and performance evaluation.

Medical Scribe

July 2023 – August 2024

Memorial Healthcare | Owosso, MI

- Documented ED and ICU encounters in MEDITECH for emergency and internal medicine physicians.
- Synthesized patient histories, labs, and assessments for up to 20 patients per shift.
- Interpreted laboratory values to streamline documentation and highlight critical findings.
- Reviewed and abstracted data from medical charts to support clinical decision-making.
- Coordinated with nursing staff to support unit operations.

Volunteer Staff

September 2022 – April 2023

Sparrow Hospital | Lansing, MI

- Supported nursing staff in an inpatient unit by providing basic care to up to 40 patients per shift.
- Measured vital signs, transported patients within the hospital, and observed diagnostic imaging exams.

Selected Projects

[The Social Determinants of EHR Quality in US Hospitals](#)

January 2024 – April 2024

Department of Sociology, Michigan State University | East Lansing, MI

- Launched and presented a research project investigating the social determinants of electronic health record quality; conducted literature review, and compiled and analyzed complex health datasets.

[Introduction to Biostatistical Machine Learning](#)

April 2024

Department of Epidemiology & Biostatistics, Michigan State University | East Lansing, MI

- Synthesized classical biostatistical methods and modern ML in diagnostics and epidemiology.

[AI and Biomedicine: An STS Perspective](#)

December 2023

Department of Sociology, Michigan State University | East Lansing, MI

- Analyzed AI in biomedicine through a socio-technical lens, focusing on EHRs as systems that shape data quality, clinician behavior, and downstream AI performance.

[Prospect of Machine Learning in Medical Microbiology](#)

November 2022

Department of Microbiology, Genetics & Immunology, Michigan State University | East Lansing, MI

- Reviewed AI applications in microbiology with emphasis on workflow-aligned deployment strategies.

Education

Bachelor of Science in Human Biology, Minor in Bioethics
Michigan State University | East Lansing, MI

September 2020 - May 2024
GPA: 3.91 (Honors)

Awards and Honors

Distinguished Freshman Scholarship
Michigan State University Honors College

Issued September 2020

- Full-tuition academic scholarship awarded for outstanding academic achievement.

Technical Skills

- **Programming:** JavaScript, TypeScript, Python, C++, Bash, SQL, R
- **Web Development:** React, Next.js, Node.js, Flask, FastAPI, Prisma, PostgreSQL, Vite, HTML/CSS, TailwindCSS
- **Artificial Intelligence:** PyTorch, PyTorch Geometric, scikit-learn, Transformers, Graph Neural Networks (GNNs), Multilayer Perceptrons (MLPs), Convolutional Neural Networks (CNNs), Mixture of Experts (MoE), Transfer Learning (TL), Large Language Models (LLMs), Prompt Engineering, Hyperparameter Tuning, Model Evaluation Metrics, OpenAI API
- **Tools & Platforms:** Git/GitHub, Linux, Vercel, Cloudflare, CLI, Jupyter Notebook
- **Data Analysis:** NumPy, Pandas, SciPy, Matplotlib, Seaborn, scikit-learn

Languages

- **English** (Native or bilingual proficiency)
- **Spanish** (Elementary proficiency)